

UNIT I - Answer Key PDF

Problem Solving Techniques: Functions, Data Abstraction, Scoping, Algorithmic Strategies

Teacher-only answer key. Do not share this PDF with students before results are published.

Section A - MCQ Answer Key

Question	Answer	Question	Answer	Question	Answer
Q1	A	Q2	B	Q3	B
Q4	C	Q5	B	Q6	A
Q7	B	Q8	A	Q9	B
Q10	B	Q11	B	Q12	A
Q13	B	Q14	A	Q15	B
Q16	B	Q17	B	Q18	A
Q19	B	Q20	A	Q21	A
Q22	C	Q23	B	Q24	A
Q25	B	Q26	B	Q27	B
Q28	A	Q29	A	Q30	A

Section B - Fill-up Answer Key

Q.No	Accepted Answer(s)
1	task
2	function
3	parameters
4	arguments
5	recursive
6	Type
7	implementation details
8	constructors
9	selectors
10	brackets
11	scope
12	lifetime
13	mapping
14	Built-in
15	global
16	finiteness
17	posteriori
18	space
19	unambiguous
20	tradeoff

Section C - Simple Coding Answer Key

Q1. Write `square(x)` that returns the square of `x`.

```
def square(x):  
    return x * x
```

Evaluation notes: Full marks if function returns `x*x` for positive, zero, and negative integers.

Q2. Write recursive `factorial(n)`, assuming `n >= 0`.

```
def factorial(n):  
    if n == 0:  
        return 1  
    return n * factorial(n - 1)
```

Evaluation notes: Must include a base case and recursive case.

Q3. Write `is_positive_square(x)` that returns `x*x` only when `x > 0`, else `None`.

```
def is_positive_square(x):  
    if x > 0:  
        return x * x  
    return None
```

Evaluation notes: Accept equivalent conditional expression.

Q4. Write `makecity(name, lat, lon)` using a dictionary.

```
def makecity(name, lat, lon):  
    return {'name': name, 'lat': lat, 'lon': lon}
```

Evaluation notes: Keys must be `name`, `lat`, and `lon`.

Q5. Write `getname(city)`, `getlat(city)`, and `getlon(city)`.

```
def getname(city):  
    return city['name']  
  
def getlat(city):  
    return city['lat']  
  
def getlon(city):  
    return city['lon']
```

Evaluation notes: Selectors should return corresponding dictionary fields.

Q6. Write `make_pair(a, b)` that returns `[a, b]`.

```
def make_pair(a, b):  
    return [a, b]
```

Evaluation notes: Must preserve order.

Q7. Write `local_value()` that creates local `a = 7` and returns it.

```
def local_value():  
    a = 7  
    return a
```

Evaluation notes: Variable must be local to the function.

Q8. Write `shadow_value()` using local `a=7` while global `a=10` remains unchanged.

```
a = 10  
def shadow_value():  
    a = 7  
    return a
```

Evaluation notes: Do not use `global` keyword or modify global `a`.

Q9. Write `steps_to_square(x)` returning the algorithm steps list.

```
def steps_to_square(x):  
    return ['get input', 'calculate square', 'display result']
```

Evaluation notes: Return exactly the expected list.

Q10. Write `space_factor(fixed, variable)` returning `fixed + variable`.

```
def space_factor(fixed, variable):  
    return fixed + variable
```

Evaluation notes: Accept integer or numeric addition.